QA PORTFOLIO CASE STUDY

Toolshop Practice Site

Cross-browser functional, responsive and automation QA of the intentionally buggy Toolshop build

SYSTEM UNDER TEST	`with-bugs.practicesoftwaretesting.com` — Toolshop v5.0, a shop built specifically for testers to practise on
REFERENCE BUILD	`practicesoftwaretesting.com` — the same application without planted bugs
SCOPE	Navigation, catalogue, search and sort, cart action, contact, login and registration across six browser and device profiles
DELIVERABLES	This report, plus a Playwright + TypeScript regression suite with one re-runnable test per defect

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DATE

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> Portfolio QA case study on a public practice system - not client work.

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1 Executive summary

Release assessment: BLOCK

BUG-07 prevents the site's only contact and lead form from being submitted. Four High-severity defects also break primary journeys: mobile catalogue discovery, cart feedback, primary navigation and category routing.

Most findings reproduce across all six profiles; three are environment-specific and are labelled accordingly. Each confirmed defect has a focused Playwright regression test, and the final matrix completed with zero unexpected outcomes.

10
confirmed findings

+ 6 additional observations (unranked)

Findings by severity



Top issues

BUG-07	CRITICAL	Contact form — submission blocked: “Subject is required” persists after a subject is selected
BUG-16	HIGH	Catalogue filters (mobile) — the “Filters” toggle does nothing below 768 px, so search, sort and filters are unreachable
BUG-05	HIGH	Product page — “Add to cart” shows an error toast although the item is added

Test run

228	96	94	0	14.3 min
test executions	checks passed	known-bug checks failed as recorded	unexpected results	full matrix runtime

38 executions were skipped by design: desktop-only suites on the other profiles, and sort on phones, which BUG-16 blocks.

Coverage

Desktop Chrome · Desktop Safari (WebKit) · Desktop Firefox · iPhone 14 · Pixel 7 · iPad (gen 7)

2 Scope and methodology

IN SCOPE	OUT OF SCOPE
Home, category pages, product page, search, sort, cart action, main navigation, contact, login, register	Checkout and payment, account area, rentals detail, admin
Responsive layout from 360 to 1440 px, broken images, long content, unusual characters (Arabic, emoji)	Load, security and penetration testing
	Manual screen-reader pass (noted as a follow-up)

Methodology

Expected behaviour was taken from the reference build - the same application without planted bugs. Business-critical journeys (navigation -> product -> add to cart, and the contact form) received a full pass in all six profiles. Engine-independent checks - metadata, link crawl, automated accessibility checks and the breakpoint sweep - ran once in desktop Chrome.

Every defect was reproduced manually and compared with the reference build before filing. Automated or AI-assisted results that did not reproduce were discarded (see section 9). Each confirmed defect has its own re-runnable regression test.

3 Severity definitions

SEVERITY	DEFINITION
CRITICAL	Blocks a core user journey or the launch for all users.
HIGH	Breaks a primary journey or affects every visitor, or creates SEO or legal risk.
MEDIUM	Degrades the experience for some content, users or environments.
LOW	Polish, consistency or best-practice issue.

4 Test environment

PROFILE	ENGINE AND VERSION	OS	VIEWPORT	TYPE
Desktop Chrome	Chromium 153.0.8010.12 (Playwright 1.63.0)	macOS 27.0.1	1280×720	Real engine
Desktop Safari	WebKit 26.6 (Playwright build)	macOS 27.0.1	1280×720	Emulated
Desktop Firefox	Firefox 155.0	macOS 27.0.1	1280×720	Real engine
iPhone 14	WebKit 26.6	iOS profile	390×664	Emulated
Pixel 7	Chromium 153	Android profile	412×839	Emulated
iPad (gen 7)	WebKit 26.6	iPadOS profile	810×1080	Emulated

- Desktop Safari runs Playwright’s WebKit build, not Apple’s Safari binary.
- Evidence screenshots were taken at 1440×900 in desktop Chrome unless stated otherwise.
- Three defects are environment-specific: **BUG-13** (Chromium only), **BUG-14** (Firefox only) and **BUG-16** (below 768 px, also confirmed in a resized desktop Chrome window).

5 Findings overview

ID	FINDING	SEVERITY	WHERE
BUG-07	Contact form — submission blocked: “Subject is required” persists after a subject is selected	CRITICAL	All profiles
BUG-16	Catalogue filters (mobile) — the “Filters” toggle does nothing below 768 px, so search, sort and filters are unreachable	HIGH	iPhone 14, Pixel 7; desktop Chrome at 390 px
BUG-05	Product page — “Add to cart” shows an error toast although the item is added	HIGH	All profiles
BUG-01	Main navigation — the “Home” link opens the Contact page	HIGH	All profiles
BUG-02	Categories menu — “UNDEFINED” opens an empty category and “Chainsaws” opens a 404 page	HIGH	All profiles
BUG-03	Product sort — sort labels are inverted	MEDIUM	Desktop and iPad (sort hidden on phones)
BUG-06	Contact form — an invalid email address is accepted	MEDIUM	All profiles
BUG-13	Product page — “Add to favourites” overlaps 50 px of the “Add to cart” button (Chromium only)	MEDIUM	Chromium only
BUG-09	Site header — the logo and the search icon are broken images	MEDIUM	All profiles
BUG-14	Product grid — long product names are clipped mid-word (Firefox only)	LOW	Firefox only

6 Detailed findings

BUG-07 CRITICAL

Contact form — submission blocked: “Subject is required” persists after a subject is selected

> **IMPACT** Every visitor is affected: the site’s only contact and lead form cannot be sent, so no enquiry reaches the business.

ENVIRONMENT All six profiles. Evidence: desktop Chrome 153, macOS 27.0.1, 1440×900.

STEPS TO REPRODUCE

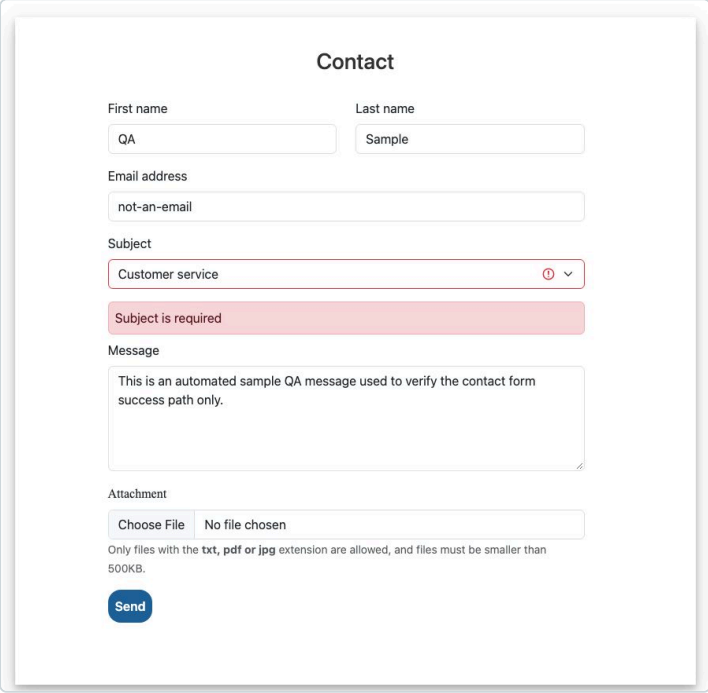
1. Open `/#/contact`.
2. Enter First name `QA`, Last name `Sample` and Email `qa.sample@example.com`.
3. Select Subject **Customer service** (also reproduced with **Webmaster** and **Payments**).
4. Enter a message of 50 or more characters.
5. Click **Send**.

EXPECTED

The message is sent and a confirmation appears, as on the reference build: “Thanks for your message! We will contact you shortly.”

ACTUAL

The Subject field turns red with “Subject is required” although its value is `customer-service`; nothing is sent and no confirmation appears.



“Customer service” is selected, yet “Subject is required” is shown. The invalid email (BUG-06) raises no error.
`evidence/BUG-06-07-contact-form.png (cropped)`

RECOMMENDATION Check that the validator reads the same form control and value the select is bound to; the form-control name or the `required` validator on `subject` is the likely cause.

BUG-16 **HIGH**

Catalogue filters (mobile) — the “Filters” toggle does nothing below 768 px, so search, sort and filters are unreachable

> **IMPACT** At phone widths, users cannot reach catalogue search, sorting or filters, blocking core product-discovery controls.

ENVIRONMENT iPhone 14 (WebKit, emulated) and Pixel 7 (Chromium, emulated). Also reproduced in desktop Chrome 153 resized to 390×844, so this is a breakpoint defect, not an emulation artefact. Not applicable at 768 px and wider, where desktop and iPad show the sidebar.

STEPS TO REPRODUCE

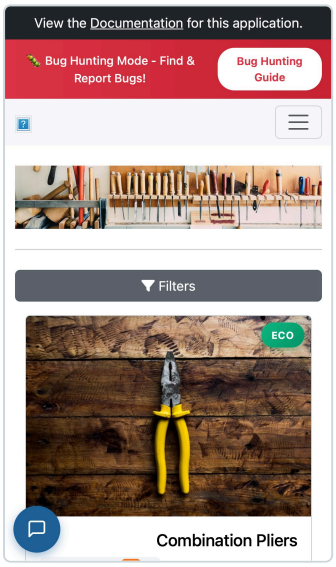
1. Open the home page at a phone width (below 768 px).
2. Tap **Filters** above the product grid.

EXPECTED

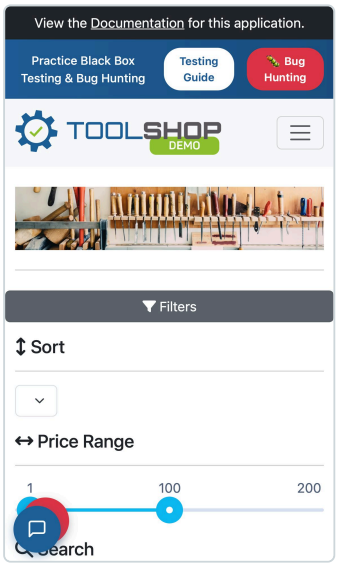
The panel expands to show Sort, Price Range, Search and the category and brand filters, as on the reference build.

ACTUAL

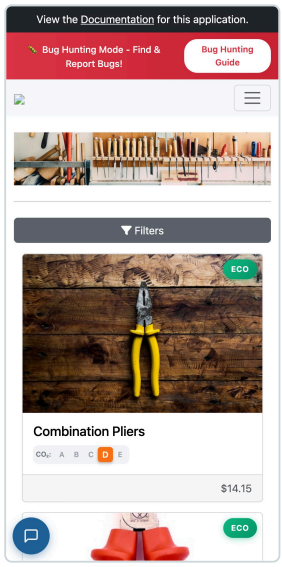
Nothing happens: the panel stays collapsed, and the search box and sort select remain hidden.



iPhone 14 (emulated), after tapping Filters: panel stays closed.
`evidence/BUG-16-filters-tapped-iphone14-emulated-bug.png`



Reference build, same step: panel opens.
`evidence/BUG-16-filters-tapped-iphone14-emulated-ref.png`



Pixel 7 (emulated), after tapping Filters: panel stays closed.
`evidence/BUG-16-filters-tapped-pixel7-emulated-bug.png`

RECOMMENDATION The buggy build renders the toggle as an `<a>` with `aria-controls="collapseExample"`, while the reference build uses `<button data-bs-target="#filters">`. Bind the collapse handler to the `#filters` panel and make sure the anchor’s default hash navigation does not swallow the click in the hash-routed app.

BUG-05 HIGH

Product page — “Add to cart” shows an error toast although the item is added

> **IMPACT** A core commerce action reports failure, so shoppers retry (adding duplicates) or leave.

ENVIRONMENT All six profiles. Evidence: desktop Chrome, 1440×900.

STEPS TO REPRODUCE

1. Open any product, e.g. `/#/product/1` (Combination Pliers).
2. Click **Add to cart**.

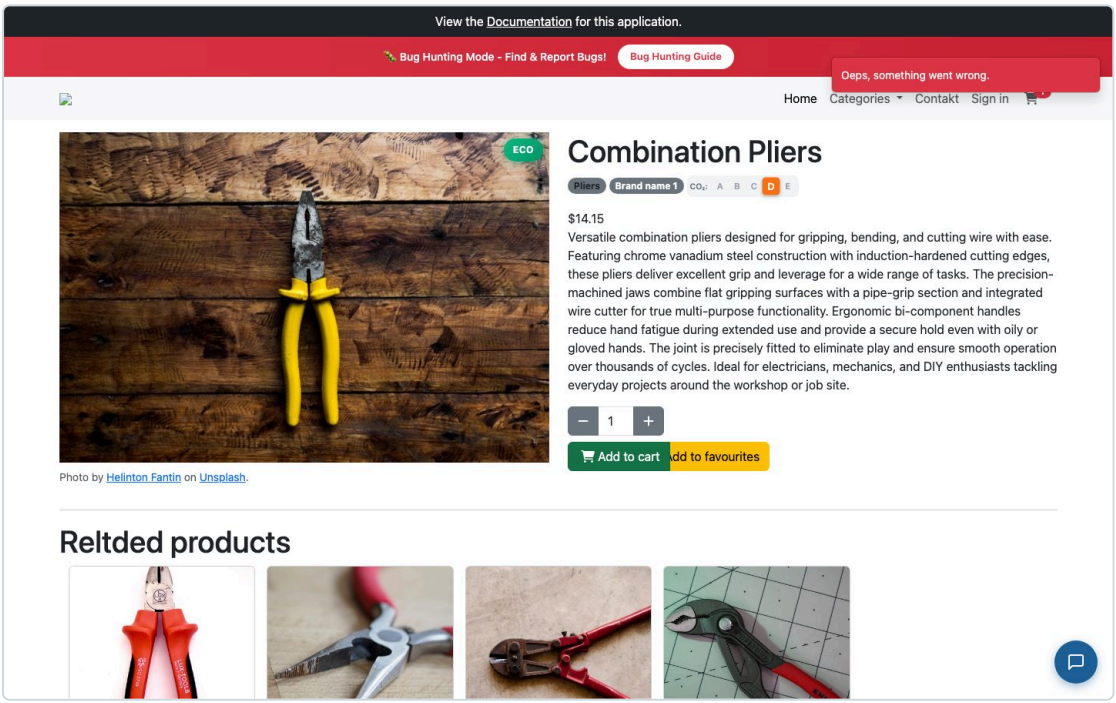
EXPECTED

A green toast reads “Product added to shopping cart.” and the cart badge shows the chosen quantity, as on the reference build.

ACTUAL

A red toast reads “Oeps, something went wrong.”, yet the cart badge still increments to 1: the item was added and only the message is wrong.

RELATED The quantity cannot be raised in any case, because the “+” button does nothing (BUG-04).



Error toast after a successful add; the cart badge shows 1.
`evidence/BUG-05-add-to-cart-error-toast.png`

RECOMMENDATION Review the success branch of the add-to-cart call; the error toast appears to fire on a 2xx response.

BUG-01 HIGH

Main navigation — the “Home” link opens the Contact page

> **IMPACT** Primary navigation on every page sends visitors to the wrong place.

ENVIRONMENT All six profiles; the link target is the same in the mobile menu.

STEPS TO REPRODUCE

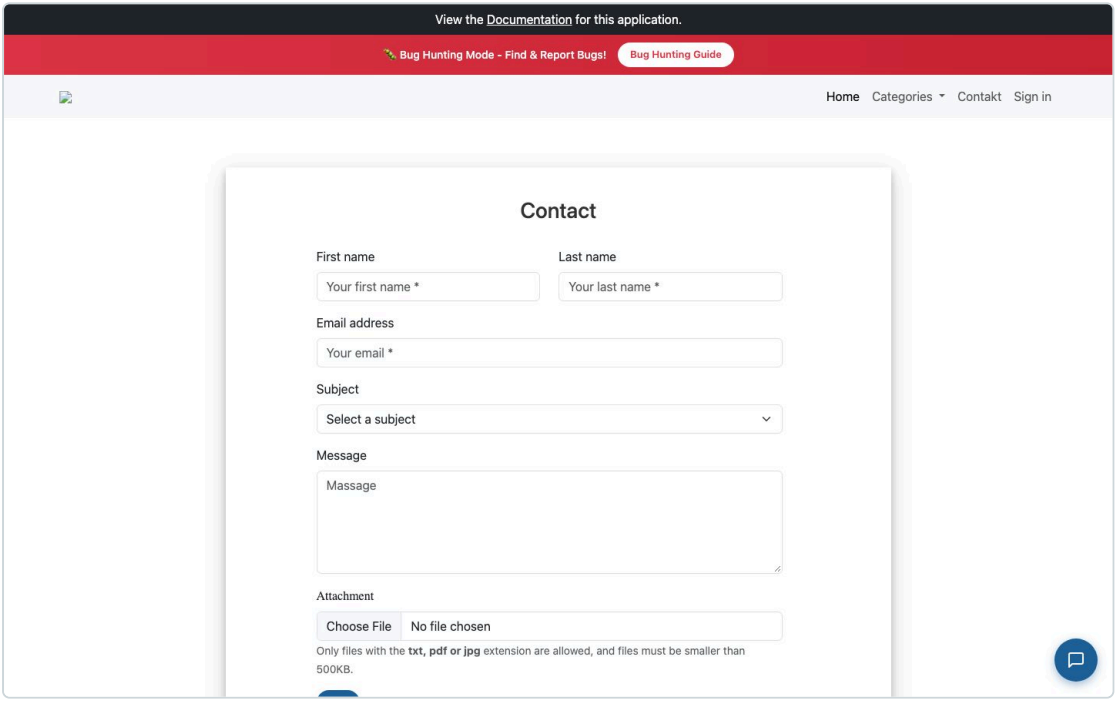
1. Open any page, e.g. a product page.
2. Click **Home** in the top navigation.

EXPECTED

The home (catalogue) page opens at `/`.

ACTUAL

The URL changes to `/#/contact` and the Contact form opens; the link’s `href` is `#/contact`.



After clicking Home: the Contact page at /#/contact.
`evidence/BUG-01-home-link-opens-contact.png`

RECOMMENDATION Point the `routerLink` of `nav-home` to `/` instead of `/contact`.

BUG-02 **HIGH**

Categories menu — “UNDEFINED” opens an empty category and “Chainsaws” opens a 404 page

> **IMPACT** Two of the four category entries are dead ends, and every visitor sees a raw “UNDEFINED” label in the menu.

ENVIRONMENT All six profiles. The link crawl (34 internal routes) found exactly these two broken routes.

STEPS TO REPRODUCE

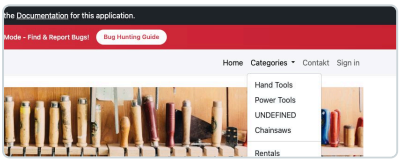
1. Click **Categories**, then **UNDEFINED**.
2. Click **Categories**, then **Chainsaws**.

EXPECTED

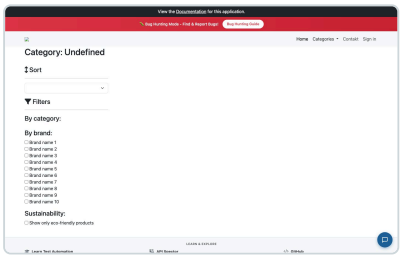
Both entries open real categories with products; the reference build shows “Other” and “Special Tools” in these positions.

ACTUAL

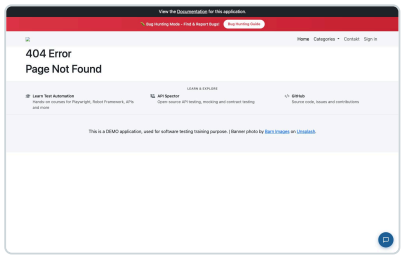
“UNDEFINED” opens `/#/category/undefined` titled “Category: Undefined” with 0 products, and “Chainsaws” opens `/#/chainsaws`, which shows “404 Error – Page Not Found”.



Categories menu with the raw “UNDEFINED” entry.
`evidence/BUG-02-categories-menu.png`



“Category: Undefined” with 0 products.
`evidence/BUG-02-category-undefined.png`



“Chainsaws” leads to a 404 page.
`evidence/BUG-02-chainsaws-404.png`

RECOMMENDATION Fix the menu item whose slug and label come from an undefined property, and add the missing `/category/` prefix to the special-tools link.

BUG-03 **MEDIUM**

Product sort — sort labels are inverted

> **IMPACT** Sorting works but does the opposite of what it says, so shoppers conclude it is broken.

ENVIRONMENT Desktop Chrome, Safari (WebKit), Firefox and iPad. On phones the sort control cannot be reached at all (BUG-16).

STEPS TO REPRODUCE

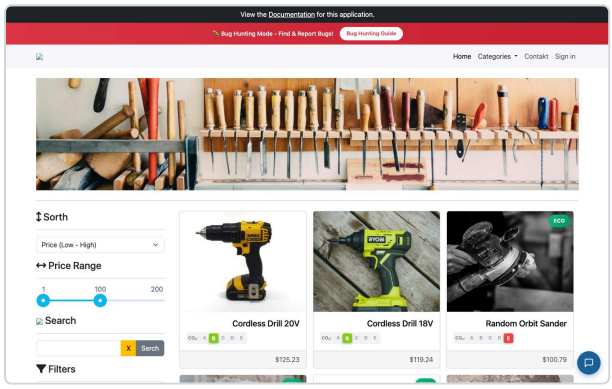
1. Open the home page.
2. Select Sort **Price (Low - High)**.
3. Select Sort **Name (A - Z)**.

EXPECTED

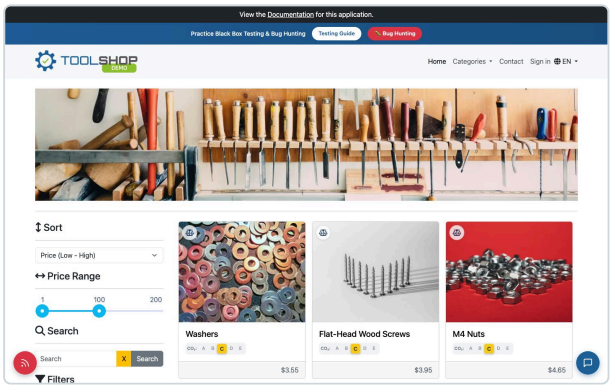
Prices are listed in ascending order ($3.55, $3.95 … on the reference build) and names from A to Z.

ACTUAL

“Price (Low - High)” lists $125.23, $119.24, $100.79 … and “Name (A - Z)” lists Wood Saw, Thor Hammer, Slip Joint Pliers …, because the option values are swapped (`price,asc` is labelled “High - Low”).



“Price (Low - High)” selected: most expensive first.
`evidence/BUG-03-sort-low-high-shows-high-first.png`



Reference build, same option: cheapest first.
`evidence/REF-sort-low-high.png`

RECOMMENDATION Swap the labels or the values of the four sort options.

BUG-06 **MEDIUM**

Contact form — an invalid email address is accepted

> **IMPACT** Unusable leads would reach the inbox. The defect is currently masked by BUG-07 and will surface as soon as BUG-07 is fixed.

ENVIRONMENT All six profiles.

STEPS TO REPRODUCE

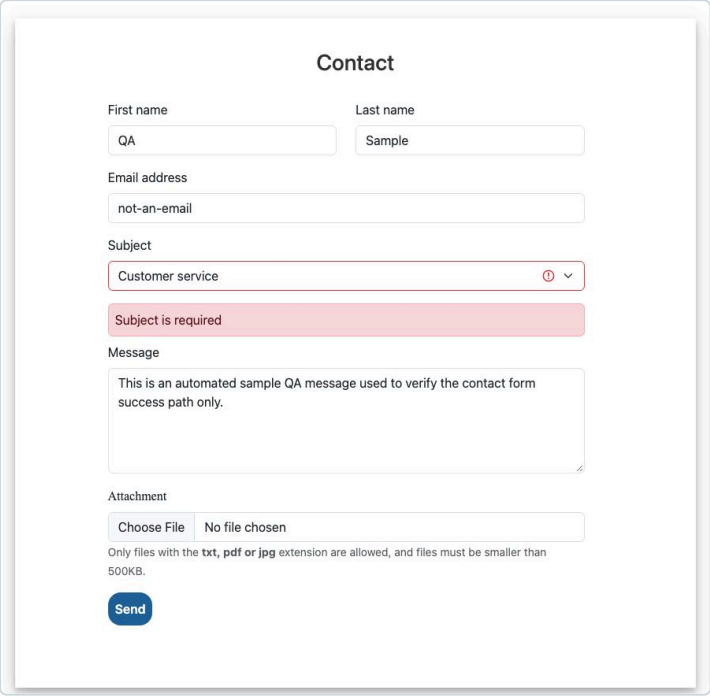
1. Open `/#/contact`.
2. Enter Email `not-an-email`.
3. Fill in the remaining fields.
4. Click **Send**.

EXPECTED

The form shows “Email format is invalid”, as on the reference build.

ACTUAL

No email error is shown; only the incorrect subject error from BUG-07 appears.



“not-an-email” in the Email field raises no error.
`evidence/BUG-06-07-contact-form.png (cropped)`

RECOMMENDATION Restore the email pattern validator on the `email` control.

BUG-13 **MEDIUM**

Product page — “Add to favourites” overlaps 50 px of the “Add to cart” button (Chromium only)

IMPACT

ENVIRONMENT Desktop Chrome 153 and Pixel 7 (Chromium, emulated). Not reproduced in desktop Safari (WebKit), iPhone 14, iPad or Firefox, where the buttons join with the normal 1 px border. Pixel 7 is emulated, so this needs confirmation on a real Android device.

STEPS TO REPRODUCE

1. Open `/#/product/1`.
2. Inspect the two buttons below the quantity field.

EXPECTED

The buttons sit side by side and the full “Add to cart” label is visible.

ACTUAL

In Chromium the favourites button has `margin-left: -50px` and covers the right-hand part of “Add to cart”; in WebKit and Firefox the computed `margin-left` is `-1px`.



Desktop Chrome: label cut to “Add to”.
`evidence/BUG-13-buttons-chrome-desktop.png`



Pixel 7 (emulated): same overlap.
`evidence/BUG-13-buttons-pixel7-emulated.png`



Safari (WebKit): no overlap.
`evidence/BUG-13-buttons-safari-desktop-webkit.png`

RECOMMENDATION Find and remove the Chromium-targeted CSS rule, or the browser check, that sets the negative margin.

BUG-09 **MEDIUM**

Site header — the logo and the search icon are broken images

> **IMPACT** The brand logo is missing on every page, and each page load fires two 404 requests and console errors.

ENVIRONMENT All six profiles.

STEPS TO REPRODUCE

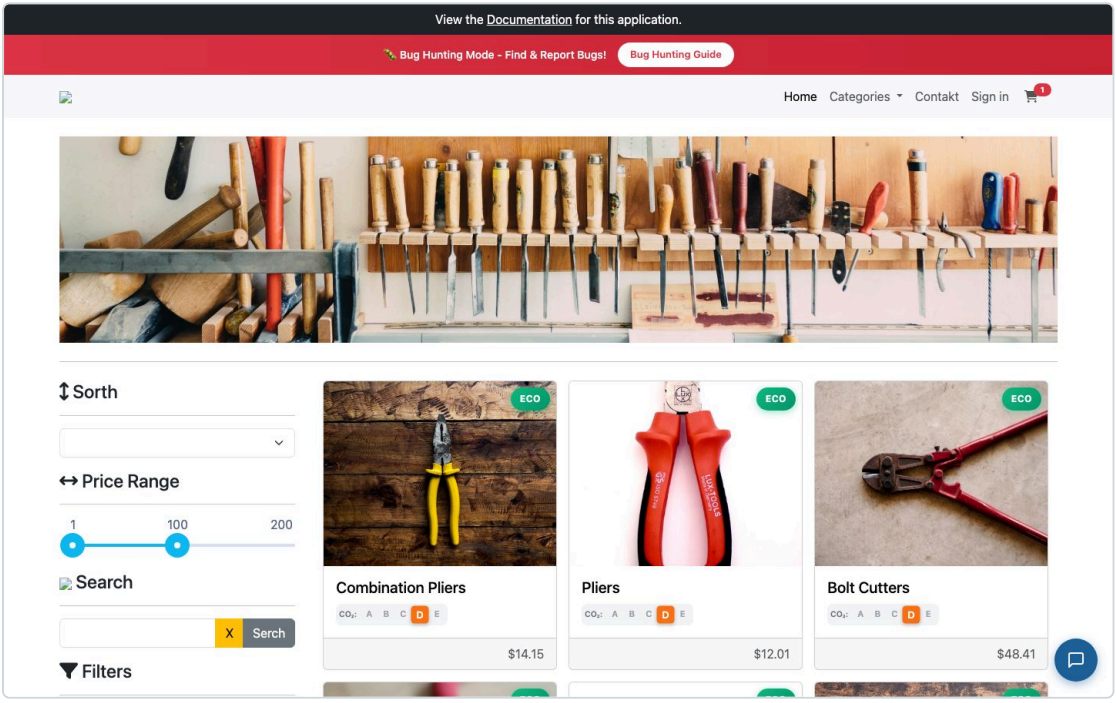
1. Open the home page with DevTools on the **Network** tab.
2. Inspect the logo and the search icon.

EXPECTED

The logo and the search icon render.

ACTUAL

Both `<img>` elements point to `broken.png`, which returns HTTP 404, so the browser shows the broken-image icon (top-left logo and next to “Search”); neither image has `alt` text (see the accessibility audit).



Broken-image icons in place of the logo and the search icon.
`evidence/BUG-09-broken-images-home.png`

RECOMMENDATION Restore the asset paths and add `alt` text (or `alt=""` for the decorative icon).

BUG-14 LOW

Product grid — long product names are clipped mid-word (Firefox only)

> **IMPACT** Shoppers cannot read the full product name (“Claw Hammer with Shock R”) on the listing; there is no ellipsis or tooltip.

ENVIRONMENT Firefox 155 desktop only. Chromium and WebKit wrap the name onto two lines, and the reference build is correct in Firefox.

STEPS TO REPRODUCE

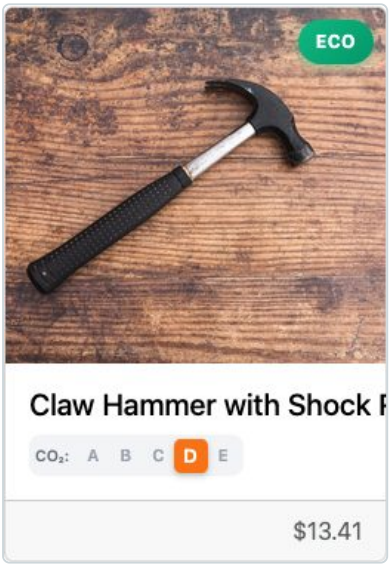
1. Open the home page in Firefox.
2. Find the product “Claw Hammer with Shock Reduction Grip”.

EXPECTED

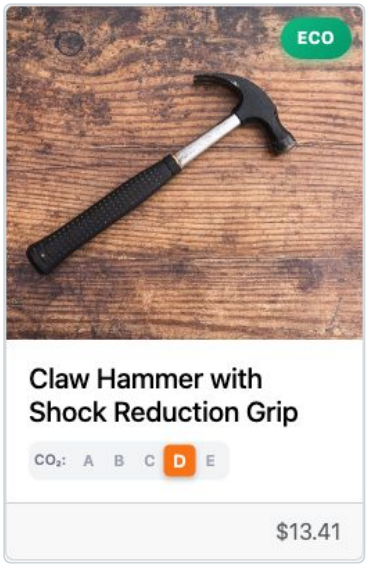
The full name is visible, wrapped onto two lines as in Chrome.

ACTUAL

The name stays on one line (`white-space: nowrap`) — 370 px of text in a 225 px box — and is clipped without an ellipsis.



Firefox: name clipped mid-word.
`evidence/BUG-14-long-name-firefox.png`



Chrome: name wraps correctly.
`evidence/BUG-14-long-name-chrome.png`

RECOMMENDATION Remove the Firefox-targeted `white-space: nowrap` rule, or add `text-overflow: ellipsis` with a `title` attribute.

7 Additional observations

Lower-priority defects, not ranked by severity. Each one has its own test in the regression suite.

ID	AREA	OBSERVATION
BUG-04	Product page	The quantity “+” button does nothing.
BUG-08	Contact form	The subject list contains “Error 101: Subject not found” and “Error 202: Translation error”.
BUG-10	Product page	The related-products heading reads “Reltded products”, and it is a second `<h1>`.
BUG-11	Login	An empty submit shows “Invalid email or password” instead of field-level “required” messages.
BUG-12	Copy	Typos: “Contakt”, “Sorth” and “Serch”.
BUG-15	Product grid	Product names mix left and right alignment.

8 Automated audit highlights

Report-only suites: their results inform the backlog but are not filed as individual defects.

Metadata (4 key pages)	Every page is missing a meta description and a canonical URL, none has an `<h1>`, and all pages share one `<title>` (“Practice Software Testing - Toolshop - v5.0 with bugs”). Full list: `audit/seo-findings.json`.
Accessibility - automated axe-core checks (WCAG 2.2 A/AA rules)	`image-alt` (critical; 10–11 product images per listing page), `select-name` (unlabelled sort select), `label` (unlabelled field on the home page), `button-name` (icon-only button on login), `color-contrast` (serious; 2–3 elements per page) and `list` (invalid list markup). Full list: `audit/a11y-findings.json`. An automated scan covers only part of WCAG; a keyboard and screen-reader pass is still needed.
Responsive sweep (360–1440 px; home, contact, login)	No horizontal overflow, and key elements are visible at every breakpoint. Screenshots are in the HTML report.
Links	34 internal routes crawled (`audit/links-findings.json`); only the two in BUG-02 are broken. Eleven external Unsplash credit links return 401 to automated requests (bot protection), so they are marked “unverified” rather than broken.

9 AI-assisted workflow and validation controls

> **USED FOR**
>
> AI accelerated test-permutation design, initial Playwright scaffolding, console/network parsing and reference-build comparison. No finding was accepted from AI or automation output alone; each reported defect was independently reproduced and verified.

Validation controls and corrections

1. **False positive — search.** The first automated search test reported “Bolt Cutters” among the results for “pliers”. Re-running by hand showed the result list was correct: the test read the grid before it re-rendered. Fixed with a poll on the result list; not filed.
2. **False positive — link crawl.** The first crawl marked all 32 pages as broken. Hash-route navigations return no HTTP response object, which the script treated as an error, and a loose “404” text match hit normal pages. The rewritten detector looks for the actual 404 view and “undefined” titles, and now finds exactly the two real broken routes.
3. **Wrong scope — button overlap.** First noted as a general layout bug from a Chrome screenshot. Running the matrix showed it is Chromium-only, which changes the fix and who is affected; it is reported that way (BUG-13).
4. **Real bug first dismissed as a test issue — mobile search.** On phones the search test timed out, and the first read was “test problem: the search box sits in a collapsed panel”. Opening the panel by hand worked on the reference build; on the buggy build the toggle did nothing, and the same happened in a resized desktop Chrome window. That became BUG-16, the most important mobile finding.
5. **Not a bug — external links.** Unsplash links returning 401 looked broken to the script but work in a browser (bot protection). Marked “unverified”.

Every defect in section 6 was reproduced manually, step by step, and compared with the reference build before filing.

10 Regression suite

The regression suite is a Playwright + TypeScript project in this repository; the README explains setup and the “known bugs” list.

Each confirmed defect has its own focused test, marked with `knownBug('BUG-xx', …)` and expected to fail on the buggy build, so the run stays green. Run the full matrix with `npx playwright test` and open the results with `npx playwright show-report`.

When a fix lands, Playwright reports “expected to fail, but passed” for the matching test: the signal to verify the fix and delete the marker, which leaves a permanent regression check.